Spring MVC Security - Display User ID and Roles - Overview and Coding

In this lesson, we are going to explore how to display user IDs and roles within an application using Spring Security. This can be vital for personalizing user experiences and providing appropriate access controls.

So here's what we are aiming for: On our homepage, we will leverage Spring Security to show not only the user ID but also the roles associated with that user. This functionality can provide valuable insights into who is using the system and what permissions they have.

Spring Security offers excellent support for retrieving and displaying both user IDs and roles, simplifying the integration process. Let's take a closer look at how we can accomplish this.

Basic Development Process

1. **Display the User ID**: Using the expression `sec.authentication.principle.username`, we can fetch and display the user ID of the currently logged-in user.
2. **Display the User Roles**: By employing `sec.authentication.principle.authorities`, which is synonymous with user roles, we can render the roles assigned to the user.

That's the core concept. Now, we'll dive into the code and test it out in our integrated development environment (IDE).

Displaying the User ID

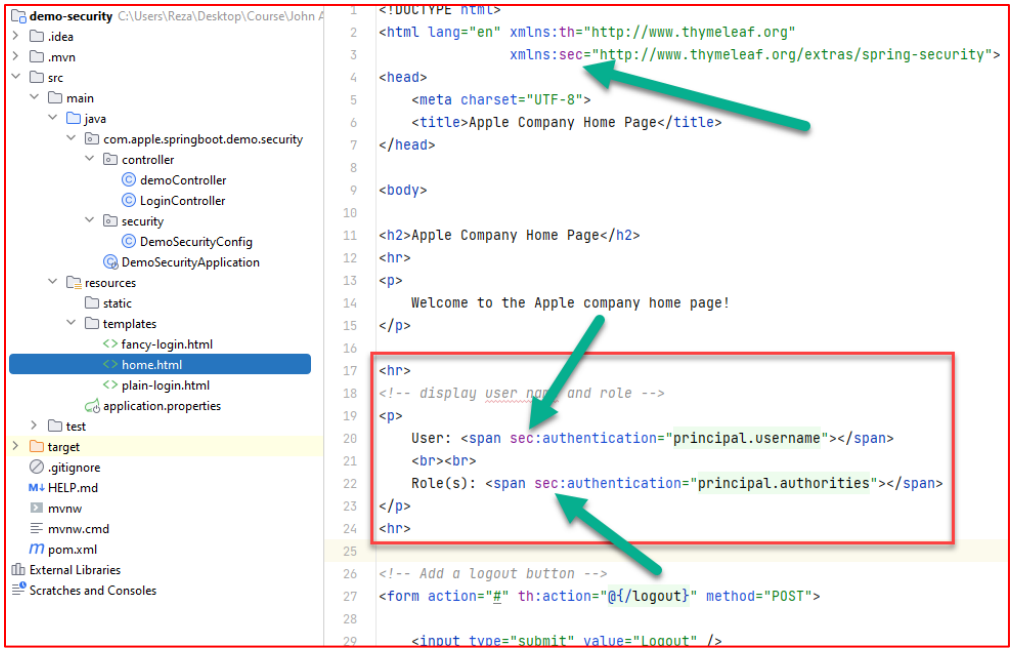
Now, let's move on to the coding part. We'll make some quick annotations in `home.html` and lay down horizontal rules to ensure everything remains organized.

Add a useful comment for future reference: "`<!-- display username and role -->`".

To retrieve the User ID, we'll use the expression `sec:authentication`, and the property will be set to `principle.username`. This expression will help us get the User ID for the logged-in user.

We'll also add some line breaks (
 tags) to create a neat visual separation.

Next, for displaying the roles, we'll use `sec:authentication` with the property `principle.authorities`. Remember, "authorities" is equivalent to "roles", and it will allow us to display the list of roles for the user.



Testing the Implementation

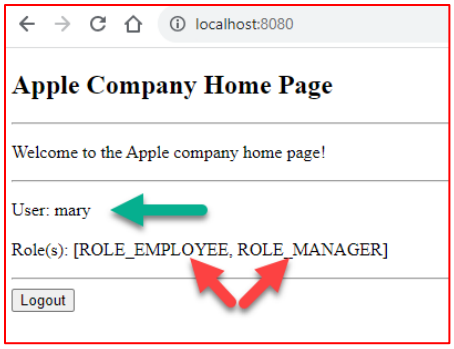
Let's run our application, hoping for the best!

Our app is up and running. After logging in with the credentials "John" and "test123", we can see the username and the role, which appears as "role_employee".



Do note that Spring Security uses the "role_" prefix for roles by default. So even if our source code has roles like admin, employee, and manager, they'll appear with this prefix. But this behavior is customizable, and we'll explore that in a later video.

We can also log out and log in as another user, such as "Mary", to see the roles dynamically adjust based on the user.



Summary

We've successfully implemented the functionality to display the User ID and roles using Spring Security. This will provide a solid foundation for managing users and roles in our application.